

```
int main()
{
    int n, tq, completed = 0, completionTime = 0;

    printf("Enter number of processes: \n");
    scanf("%d", &n);

    printf("Enter Time Quantum: \n");
    scanf("%d", &tq);

    int at[n], bt[n], rem[n], tat[n], wt[n];
    float avgtat = 0, avgwt = 0;

    // Input
    for (int i = 0; i < n; i++)
    {
        printf("Enter AT and BT for P%d: \n", i + 1);
        scanf("%d %d", &at[i], &bt[i]);
        rem[i] = bt[i];    // Remaining time
    }

    // Round Robin Logic
    while (completed < n)
    {
        int idle = 1;

        for (int i = 0; i < n; i++)
        {
            if (at[i] <= completionTime && rem[i] > 0)
            {
                idle = 0;

                if (rem[i] > tq)
                {
                    completionTime += tq;
                    rem[i] -= tq;
                }
                else
```

```

for (int i = 0; i < n; i++)
{
    if (at[i] <= completionTime && rem[i] > 0)
    {
        idle = 0;

        if (rem[i] > tq)
        {
            completionTime += tq;
            rem[i] -= tq;
        }
        else
        {
            completionTime += rem[i];
            tat[i] = completionTime - at[i]; // Turnaround Time
            wt[i] = tat[i] - bt[i]; // Waiting Time
            rem[i] = 0;
            completed++;
        }
    }
}

if (idle)
{
    completionTime++; // CPU idle
}

// Calculate averages (OUTSIDE loop)
for (int i = 0; i < n; i++)
{
    avgtat += tat[i];
    avgwt += wt[i];
}

printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);
printf("Average Waiting Time: %.2f\n", avgwt / n);

return 0;

```

```
a.out fcf.c first.sh ntiha.sh script.awk
backup fcfs.c nithya.sh prod.c
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ vi roundro.c
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ gcc roundro.c
roundro.c: In function 'main':
roundro.c:68:1: error: expected declaration or statement at end of input
 68 | }
    | ^
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ vi roundro.c
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ gcc roundro.c
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ ./a.out
Enter number of processes:
7
Enter Time Quantum:
2
Enter AT and BT for P1:
12
20
Enter AT and BT for P2:
28
51
Enter AT and BT for P3:
43
26
Enter AT and BT for P4:
71
31
Enter AT and BT for P5:
32
14
Enter AT and BT for P6:
13
42
Enter AT and BT for P7:
10
61

Average Turnaround Time: 163.86
Average Waiting Time:128.86
```